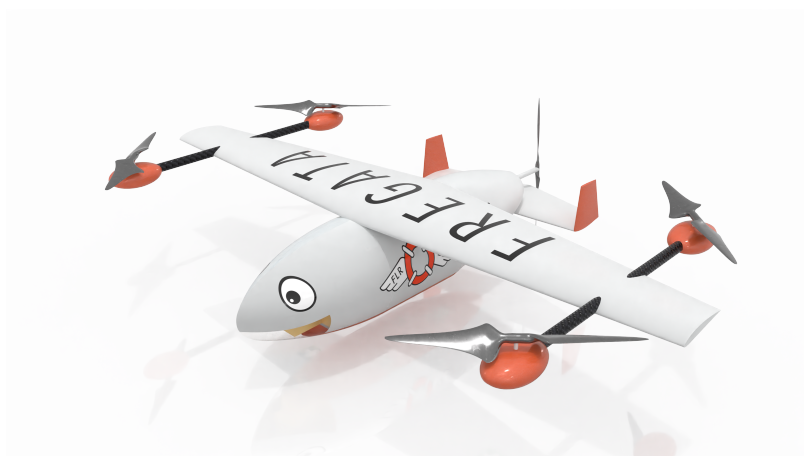




Fregata

Structural Design of Fuselage, Payload Delivery Door and Camera Cut-out with Manufacturing Processes and Materials considerations



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Abstract

This report presents a detailed design procedure for the fuselage structure of Fregata, including stress and bending moment analysis, buckling analysis and a full design of the heavy frames supporting the main components. The manufacturing procedure is described and is found to be the biggest constraining factor in the design, making the structure over-designed. Additional analysis is made on different secondary structures such as the payload door and the camera cutout. A Material selection is conducted for replaceable parts used throughout the design.

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Nomenclature

Symbols

A	Area
c	Chord Length
c	Cost
D	Diameter
E	Young's Modulus
H	Height
I	Second Moment of Area
K_{IC}	Fracture Toughness
L	Length
M	Moment
n	Loading Factor
q	Shear Flow
SF	Shear Force
t	Thickness
W	Specific Energy Absorption
W	Weight
x	Longitudinal Position
y	Lateral Position

Greek Letters

ϵ	Strain
ν	Poisson's Ratio
ρ	Density
σ	Stress
τ_s	Shear Stress

Abbreviations

CAD	Computer Aided Design
$CFRP$	Carbon Fiber Reinforced Polymer
FEA	Finite Element Analysis

FOV	Field of View
MIG	Metal Inert Gas
$MTOW$	Minimum Take Off Weight
OEI	One Engine Inoperative
PE	Polyethylene
PE	Polystyrene
SAR	Search and Rescue
UAV	Unmanned Aerial Vehicle
$VTOL$	Vertical Take-Off and Landing

Subscripts

ac	Aerodynamic Center
Al	Aluminium
b	Buckling
c	Compression
cg	Center of Gravity
$crit$	Critical
eff	Effective
fs	Front Spar
fus	Fuselage
max	Maximum
P	Tangential Component
Q	Radial Component
req	Required
rs	Rear Spar
s	shear
T	Moment Component
t	Tension
w	Wing
y	Yield

1 Introduction

From the briefing [13], different specifications are given, set as a goal for the design. The UAV needs to be able to rapidly deliver flotation and aid to a person at sea. It needs to locate and reach individuals offshore in a general area. It has to maximise the survivability in the water until other Search and Rescue (SAR) vessels can arrive. Range and speed need to be chosen carefully in consequence. The UAV should be able to operate in all weather with no deployment or recovery overhead beyond equipment readily available on rescue vessels. The aircraft must also be compact and cost-effective.

The solution proposed is an aircraft that makes coast guard operations easier and more efficient by delivering in less than 30 minutes after the alert flotation devices contained within the fuselage. The payload consists of either a set of 24 buoys or a full-size life raft for up to 6 persons. The final design of the aircraft is a fixed-wing UAV, featuring VTOL capabilities for efficient take-off and landing. The aircraft is alimented by 4 electric engines for stationary operations and a 2-stroke internal combustion engine for cruise. Landing safety is ensured by the use of an airbag when landing in a rough sea on a hard deck. The mission profile consists of a vertical take-off from one of the coast guard vessels, a 70 km cruise to reach the incident site followed by 10 minutes of loiter to locate and help the people in distress. The aircraft then either goes back to base or lands on water or the nearest vessel if the mission extends over 70 km.

The project team is composed of 22 individuals, split into 6 different subgroups with specific subjects to work on. Great challenges arise when collaborating between lots of teams, ensuring communication to avoid wasted work. This report focuses on the structural design of the UAV or more specifically on the design and sizing process of the fuselage structure, the manufacturing processes involved, the material selection for replaceable parts and the design of the secondary structures.

2 First Design Iteration

As a way to start the GDP project, it is decided to separate the group into 4 sub-teams, working on their best solution to the problem. A week time is given to do an initial sizing of the design. The first challenge is to select the right payload and then design the UAV around it. While the aim of the project is fairly wide, a focus is made on the migrant crisis in the Mediterranean Sea, representing more than 90% of the deaths by drowning [8]. In an effort of weight reduction, it is decided to use a slingshot to launch the drone. Although it adds the need for overhead equipment and a considerable constraint on the stall speed, it avoids the use of VTOL technology, massively increasing the price and weight of the UAV. This also meant that internal combustion can be used for propulsion, offering a much better energy-to-weight ratio than batteries. Regarding the recovery method, a trade-off study is done between net catching and parachute landing with an airbag. In the end, the latter is picked but comes with a considerable waterproofing challenge due to the internal combustion engine. It is finally accepted that in the case of a failed landing, the engine would potentially be damaged. The goal is also to optimise the payload weight per person saved. The ideal solution found is a self-inflating buoy, weighting 366g, the OneUp [1]. This allows the drone to carry 20 buoys for a weight of 7.3kg resulting in a relatively light design, especially compared to a 4-6 person life raft weighting at best 8.2kg [2]. To pick the right amount of buoys, a graph of the average amount of people per boat crossing the Channel is used, shown in Figure 23. 20 buoys allow saving in most cases the majority of people on board while not carrying too much payload when there are fewer people. However, buoys do not allow people to get out of contact with water, increasing the risk of hypothermia. This is acceptable as the main focus of the UAV is the Mediterranean Sea, where hypothermia is less of a risk due to the higher temperatures of the water. A Table of the risks of hypothermia depending on the temperatures can be seen in Figure 21. Carrying a heavier payload would mean that the drone can not be launched from a slingshot, resulting in a runway take-off. Therefore, the drone would directly compete with already acquired helicopters, probably making the business plan hardly profitable. For better packing density and overall weight, a twin-boom configuration is chosen, allowing the dismantling of the rear part of the UAV and packing the fuselage in the best way possible. Initial sizing of the UAV is done using the technique taught in Aeronautical Vehicle Design Module [14]. In the end, the performance

seems fairly good, a 27kg drone, with a 250 km range and 2h worst time on target. To scale the range needed, a map surveying the incidents in the Mediterranean is used, shown in Figure 22. The range is chosen so that it can cover the majority of the zone at risk. The drone is lightweight while carrying 20 buoys, with a small consumption and easy maintenance. After a quick sizing of the components within the UAV, a CAD of the aircraft is done using Fusion 360 [15] for the presentation coming.

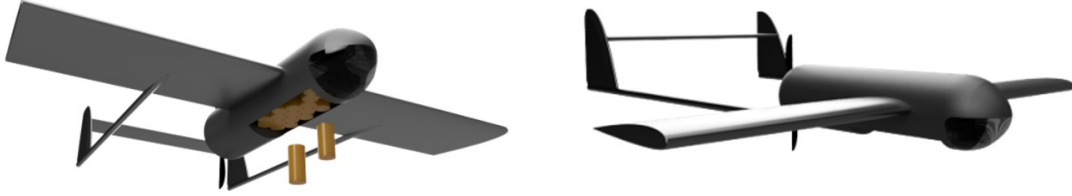


Figure 1: First UAV Concept

After the 4 presentations, a discussion between each group starts, one to one, to agree on the different pros and cons of each design, directly followed by a global debate over the final design. This is not an easy task as each team is attached to their initial idea. The first design proposed is radical compared to the others, being the only one using a slingshot and internal combustion engine. Unfortunately, due to the way the final design is chosen, by agreeing on compromised characteristics, some of the initial ideas are dissolved and the final design is much more constrained. The slingshot idea is abandoned, due to the overhead constraint. Hybrid motorisation is chosen for an electric VTOL take-off and an internal combustion engine for cruise. Nevertheless, the payload idea is used, but the sizing of it is changed to carry a small life raft for better adaptability. These choices mean a considerable increase in weight and price along with more complex controls. The new configuration however allows for an easier take-off and a better precision when dropping the payload.

3 Fuselage

The design of the fuselage requires particular attention to reach the lowest mass possible while keeping it manufacturable and structurally safe. It is therefore necessary to design a structure that can withstand the shear force and bending moment constraints along the whole fuselage. The first step is to size the different components of the UAV, giving an approximate the dimensions of the fuselage. This process is done by iteration and is refined during the project. Next, the aerodynamic and system loads acting on the fuselage are gathered from the different teams. Using the Engineering Theory of Bending, the shear force and bending moment distributions are derived. Along with a buckling analysis, the minimum skin thickness is calculated. The stringers and light frames are not required due to manufacturing constraints. This is discussed in further detail in this section. Lastly, the heavy frames are designed, made to withstand the forces coming from the main components of the aircraft acting on the fuselage, mainly the wing and tail spars.

3.1 Fuselage Sizing

The first step in the design of the fuselage is to get an estimation of the different loads and sizes. First, a discussion takes place to decide on the payload configuration. The payload gets fixed to 8.2 kg. Two payload configurations are standing out, driving the rest of the fuselage layout due to their considerable size. One has a small diameter (0.24 m) and considerable length (1.30 m). This configuration only allows to carry compact cylindrical buoys but offers a better finesse ratio. The other configuration is larger in diameter (0.36 m) but much shorter (0.95 m). This configuration however allows carrying either the buoys or the raft, greatly increasing the adaptability of the UAV. Both configurations are shown in Figures 2, 3 and 4, represented using a CAD of the payload box and its cargo.

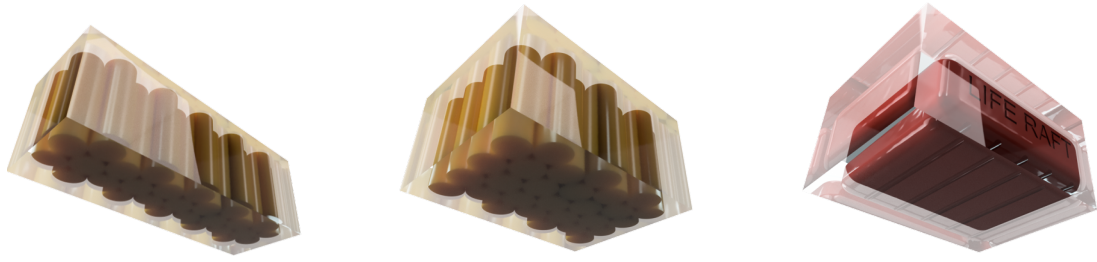


Figure 2: Payload Option 1 [1] **Figure 3:** Payload Option 2 [1] **Figure 4:** Option 2 with Raft [2]

In the end, option 2 is picked due to its versatility. The buoys are packed in cylinders, a hexagonal packing is used to achieve the best packing density possible. In collaboration with the Systems Team, initial dimensions of the most important systems are found. A first iteration of the fuselage design is developed using Fusion 360 [15] to get a proper idea of the layout and dimensions. Different persons developed a layout on their side to compare them afterwards. Two of the propositions are shown in Figure 5 and 6

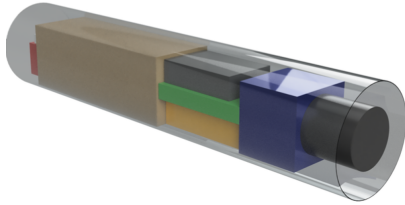


Figure 5: Thin Fuselage Configuration

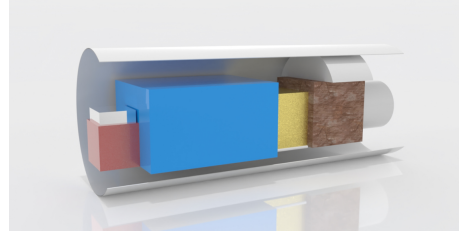


Figure 6: Wide Fuselage Configuration [3]

The initial sizing of the fuselage being done, a 3D representation of the UAV, showing the characteristics of the baseline configuration of the concept is made. Information is gathered from different teams to get an estimate of the size and placement of the wings and tail. Some renders are displayed in Figure 7

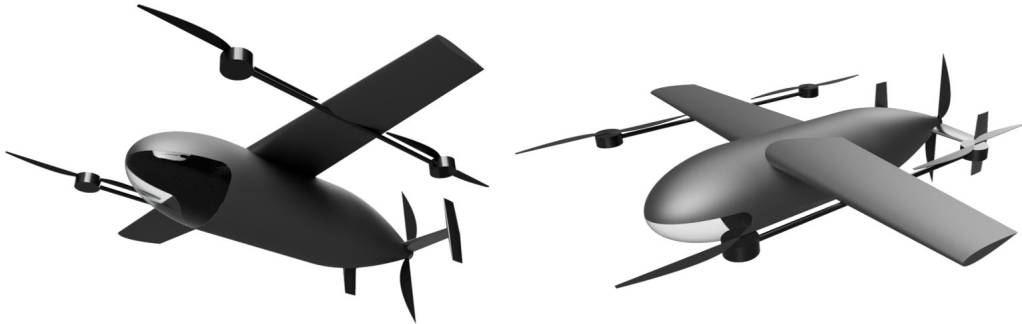


Figure 7: CAD Render

This CAD is mainly made for illustration purposes to be used as a visual verification of the correct sizing and placement of the different parts. Indeed, it shows that there is a great challenge coming from the interactions between the tail and the back VTOL propellers. This also shows that the initial values of the tail are not correct as the fuselage blocks most of the surface of the horizontal tail making control impossible.

3.2 Load Analysis

To analyse the loads, the local equilibrium of the fuselage is studied. The fuselage is simplified into a beam fully fixed at the wing spars, with each degree of freedom constrained. The reaction forces to the loads are then calculated at each spar. The loads are determined in the following way: equally spaced stations are placed at 1 cm from each other, and the fuselage weight is divided and

equally distributed across the stations. The loads from the different systems like the engine, battery or avionics are added to their respective station. The tail lift is also added to its corresponding station. The wing weight is not considered for local equilibrium. A plot of the load distribution is shown in Figure 24. The shear force distribution along the fuselage is then calculated by summing up all the loads at each station along the fuselage. Similarly, the bending moment is calculated by integration of the shear force along the fuselage.

3.2.1 First Load Case

The load case represents the UAV during a symmetric flight at a load factor of $n = 3$. A safety factor of 1.5 is considered, being the ultimate load case scenario that the fuselage would go through. The UAV is flying at manoeuvre speed at Maximum Take-off Weight. The tail lift is found to be 2274 N for this case. The shear force and bending moment distribution graphs are shown below in Figures 8 and 9 along with results for a $n=-1$ dive scenario for comparison.

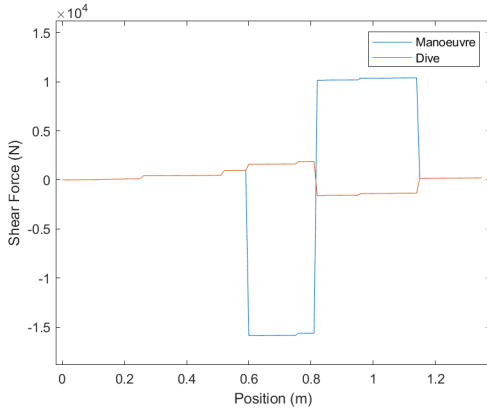


Figure 8: Shear Force Distribution

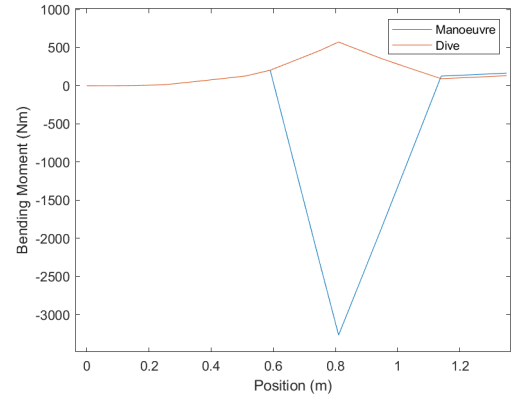


Figure 9: Bending Moment Distribution

It is clear that the shear force and bending moment induced by a $n=4.5$ manoeuvre are much greater than for a dive, even with the system loads acting in the opposite direction this is because the system loads are one order of magnitude smaller than the tail loads. It can be seen that the structure is in equilibrium as the shear force and the bending moment goes near zero at the end of the fuselage. A small error is still present due to the different approximations along with the calculation like the beam assumption for the fuselage or the tail lift force.

3.3 Other Load Cases

- One Engine Inoperative (OEI) case is typically calculated. However, in this UAV, the single engine is contained within the fuselage. Its failure does not affect the shear force and bending moment along the fuselage.
- VTOL landing method is used by the aircraft. The landing forces are absorbed thanks to the use of an airbag. The airbag being extremely soft and enveloping the fuselage, it is assumed that the load analysis is not relevant.
- The forces induced at the tail for a VTOL take-off are several orders of magnitude smaller than for manoeuvre, making the analysis of shear force and bending moment for such case not relevant.
- When grounded, the aircraft stands on its belly. However, the beam simplification does not allow to account for a point load on one small part of the fuselage radius. The analysis of a distributed load of the full weight of the drone along the beam is possible but not representative of real conditions.

Some of the cases described above are much more relevant for the design of heavy frames and will be discussed later. In summary, the maximum values are shown in Table 1 below and are used in the following steps of the design

Table 1: Maximum Loading Parameters

Max Shear Force (N)	Max Bending Moment (Nm)	Required Thickness (mm)
$1.59 \cdot 10^4$	$3.27 \cdot 10^3$	$6.72 \cdot 10^{-2}$

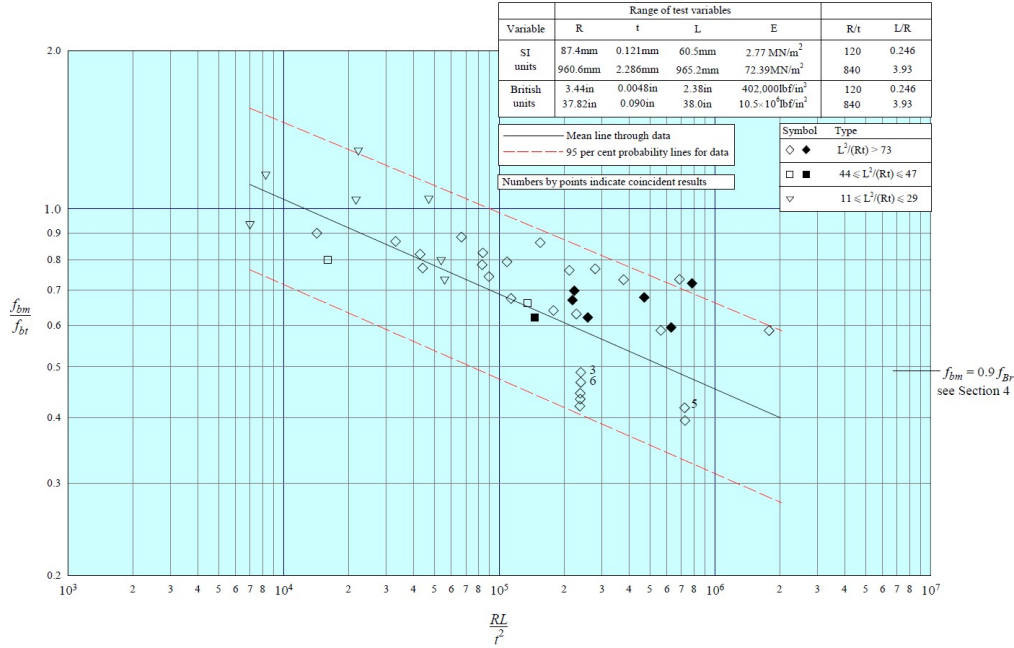
The thickness required from the max shear force is extremely small. However, manufacturing such small skin is impossible. It is, therefore, necessary to constraint thickness to 1 mm at least.

3.4 Buckling Analysis

As seen with the load analysis in the previous section, the skin thickness is more constrained by manufacturing processes rather than actual forces acting through the fuselage. This means that a monocoque design is feasible. However, it is necessary to verify the buckling performance of the fuselage. Since the panel is now a full cylinder, the methods to analyse it are more limited. Nonetheless, the ESDU 84006 [4] gives some graphs and equations to determine the risk of buckling of such a shape. A first Equation 1, shown below is given, calculating the theoretical elastic local buckling stress for a geometrically perfect cylinder loaded in axial compression.

$$f_{bt} = \frac{E}{\sqrt{3(1-\nu^2)}} \frac{t}{R} \quad (1)$$

Using Al7055, a radius of 0.177m and thickness of 1mm, the max stress found is $2.91 \cdot 10^8$. However, this equation is fairly optimistic, and a correction factor must be applied for more accurate estimation, taking into account the imperfections of a cylinder. Figure 10 shows the curve for the correction to apply depending on the cylinder parameters. A summary of the different characteristics of aluminium can be found in Figure 30.

**Figure 10:** ESDU Curve for Cylinder Buckling under Bending Load [4]

The curve is formed with experimental data, resulting in a wide spread of results. In the case of the fuselage, the length is 1.35 m. A correction factor of 0.5 is chosen to be in a conservative 95% probability regime. The buckling stress is therefore $1.46 \cdot 10^8$. The maximum stress in the fuselage is calculated using the max bending moment calculated in the previous section. Using equation 2, the maximum stress due to bending is found to be $3.39 \cdot 10^7$.

$$\frac{M_{max} R}{I_{fus}} \quad (2)$$

It is therefore clear that the fuselage is not subject to buckling as the limit stress is one order of magnitude above the maximum stress within the fuselage. Additional analysis can be done, considering the combined effect of bending and axial loading coming from drag and thrust. However, those forces are relatively small and lead to the same situation, with the limit stress being much above the maximum stress found. A graph for the correction factor to apply for axial loading is shown in Figure 25. Furthermore, cylinders are not very sensible to shear buckling so this mode and the combined ones coming along with it are not considered. Without manufacturing constraints, the minimum thickness required for buckling is calculated iteratively to be 0.49 mm.

3.5 Heavy Frame Design

It is necessary to implement heavy frames to support shear force and bending moment concentrations at critical points. The stress concentrations are located where the main components of the UAV are. Since the engine is contained within the fuselage and does not generate a dangerous amount of torque, only two sets of heavy frames are used at the wing and tail spars. The different locations of heavy frames are shown in Table 2. The overall layout of heavy frames along the fuselage is shown in Figure 11.

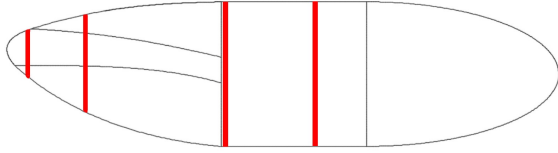


Figure 11: Heavy Frames in Fuselage Scheme

Component	Position (m)
Wing Front Spar	0.60
Wing Rear Spar	0.82
Tail Front Spar	1.17
Tail Rear Spar	1.30

Table 2: Heavy Frames Positions

It is now necessary to size the frames efficiently. First, each frame is studied using WISE curves [14]. In preliminary design, the assumption can be made that the forces and moments divide equally between the two frames of a set. A point load acting on a heavy frame can be decomposed into a radial force Q , a tangential force P and a moment T . For each of those, a WISE curve is generated with the corresponding N , M and S values, equating to the normal, tangential and moments forces in the frame of reference of the fuselage. The radial force Q is transformed in M_Q , N_Q and S_Q using equations 3, 4 and 5, where ϕ varies from 0 to 2π along the fuselage cross-section.

$$M_Q = \frac{QR}{2\pi} \left(\frac{\cos(\phi)}{2} - (\pi - \phi)\sin\phi + 1 \right) \quad (3)$$

$$N_Q = \frac{Q}{2\pi} \left(\frac{3\cos(\phi)}{2} - (\pi - \phi)\sin\phi \right) \quad (4)$$

$$S_Q = \frac{Q}{2\pi} \left(-\frac{\sin(\phi)}{2} + (\pi - \phi)\cos\phi \right) \quad (5)$$

Similarly, the tangential load P WISE curves are generated with equations 6, 7 and 8, transforming P to M_P , N_P and S_P .

$$M_P = \frac{PR}{2\pi} \left(\frac{3\sin(\phi)}{2} - (\pi - \phi)(\cos\phi - 1) \right) \quad (6)$$

$$N_P = \frac{P}{2\pi} \left(\frac{\sin(\phi)}{2} - (\pi - \phi)\cos\phi \right) \quad (7)$$

$$S_P = \frac{P}{2\pi} \left(-1 - \frac{\cos(\phi)}{2} + (\pi - \phi)\sin\phi \right) \quad (8)$$

Equally, the moment load T WISE curves are generated with equations 9, 10 and 11, transforming T to M_T , N_T and S_T

$$M_T = \frac{T}{2\pi} (\pi - 2\sin\phi - \phi) \quad (9)$$

$$N_T = \frac{T}{2\pi R} \left(\frac{3\cos(\phi)}{2} + (\pi - \phi)\sin\phi \right) \quad (10)$$

$$S_T = \frac{T}{2\pi R} (1 + 2\cos\phi) \quad (11)$$

From the WISE curves can be deducted the global axial force N , shear S force and moment M in the fuselage cross-section frame of reference by summing up the different forces calculated. This gives a global distribution of the N , S and M load and moments over the heavy frame. The reaction forces and bending moments from the wings or tail can be transformed to get a WISE curve, with overall radial, tangential and moment components independently of their position on the frame. For both the wing and the horizontal tail, the maximum shear force and bending moment are reached at a $n=4.5$ manoeuvre. Neither during a vertical take-off nor when standing on its belly does the aircraft exceeds those forces. The resulting WISE curves for both components are shown below in Figures 12 and 13.

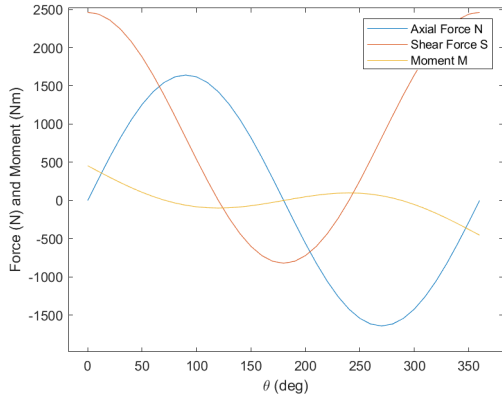


Figure 12: Wing WISE Curves

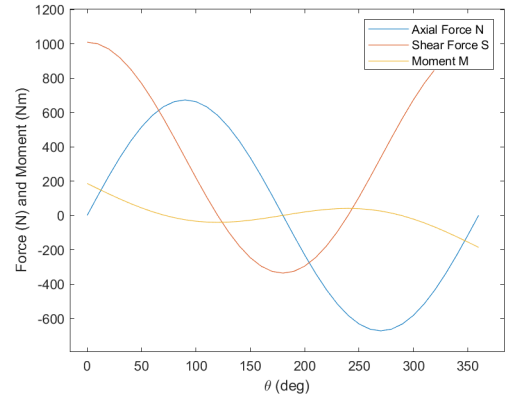


Figure 13: Tail WISE Curves

From the WISE curves, the maximal forces and moment going through the fuselage can be deducted. It is then possible to calculate the areas needed to withstand such forces by dividing respectively by the yield strength of the chosen material σ_y and the shear yield stress $\sigma_s = \sigma_y/\sqrt{3}$, as shown in the following equations.

$$A_{axial} = \frac{N_{max}}{\sigma_{yield}} \quad (12)$$

$$A_{shear} = \frac{S_{max}}{\sigma_{shear}} \quad (13)$$

Along with the constraints in area, the heavy frame cross-section also needs to satisfy the moment of inertia I_{req} needed to withstand the maximum moment along the fuselage circumference. The required value can be calculated using Equation 14, where H is the height of the frame.

$$I_{req} = \frac{M_{y_{max}} H}{2\sigma_{yield}} \quad (14)$$

For ideal structural efficiency, an I-shaped cross-section is used, outperforming rectangular or C-shaped cross-section. The moment of inertia of an I-shape beam is calculated using equation 15. It is found that the requirement on the moment of inertia is more important than the area. Using an iterative approach on MATLAB [16], the dimensions satisfying moment of inertia requirement I_{req} with the smallest area possible is found. Minimum values for thickness, height and length are set to 1mm for manufacturing considerations.

$$I = \frac{BH^3}{12} - (B-t)\frac{(H-2t)^3}{12} \quad (15)$$

Table 3 gives a summary of the main parameters for the two sets of heavy frames designed above.

Part	$A_{axial}(m^2)$	$A_{shear}(m^2)$	$I_{req}(m^4)$	$t(m)$	$H(m)$	$B(m)$
Wing	$2.43 \cdot 10^{-6}$	$6.32 \cdot 10^{-6}$	$8.74 \cdot 10^{-9}$	0.001	0.026	0.007
Tail	$1.99 \cdot 10^{-6}$	$5.18 \cdot 10^{-6}$	$2.89 \cdot 10^{-9}$	0.001	0.021	0.005

Table 3: Heavy Frames Positions

3.6 Fuselage Parameters Summary

The main parameters for the fuselage structure are summarised in Table 4. Unfortunately, the weight optimisation for the fuselage is not ideal as the sizing limits mainly come from manufacturing limitations rather than actual yield or buckling limitations. The mass increase due to these constraints is 50% or 2.22 kg. This is not optimal in a UAV where mass is such a big constraint.

Part	Dimensions (m)	Volume (m^3)	Weight (kg)
Fuselage Skin	$t=0.001$, $L=1.35$	1.64	4.44
Wing Heavy Frame	$t=0.001$, $H=0.026$, $B=0.007$	$1.67 \cdot 10^{-5}$	0.118×2
Tail Heavy Frame	$t=0.001$, $H=0.021$, $B=0.005$	$4.34 \cdot 10^{-5}$	0.045×2

Table 4: Summary of Fuselage Characteristics

4 Manufacturing

It is essential to consider the manufacturing processes behind the different design choices of the aircraft. It allows verifying that those are feasible. The different methods chosen can have a strong effect on the final price of the product, especially if the production is small. The main manufacturing processes for the aircraft structure are discussed in this section

4.1 Fuselage Skin

The fuselage can be separated into 3 different parts. The main body with constant radius and the nose and aft body with a variable radius. The main body is manufactured using aluminium extrusion to produce a constant radius aluminium cylinder. Extrusion creates lengths of aluminium of fixed cross-section by forcing a block of metal to pass through a die opening of a smaller area [17]. An illustration of the process is shown in Figure 14. This method allows for the production of well-finished and constant-thickness cylinders. The nose and aft body of the fuselage are manufactured by press moulding the shapes using an appropriate outside shape mould. Either the aluminium gets moulded using a simple hand and hammer as shown in Figure 15 or with a machine that presses the sheet with an inside mould. The cheapest choice between the two very much depends on the amount of UAV that will be produced. In a small production, the cost of developing and building a full process machine will surpass the price of labour. The trend will switch if the production gets high enough. This UAV aims to be produced in between 100 to 200 units. At such levels, a full process machine is probably the best choice. The three sections are welded together using metal inert gas (MIG) welding. This is an arc welding technique that uses a wire electrode fed in the weld with a gun [18]. The aluminium parts are melted together, forming a strong joint. Gas is used near the electrode to avoid the spread of contaminants. MIG method offers a higher depositing rate than most of the others.

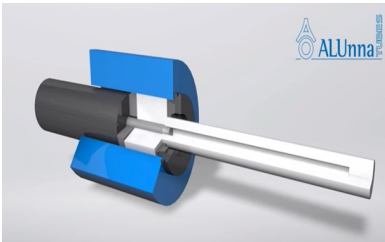


Figure 14: Aluminium Extrusion Process [5]



Figure 15: Press Moulding of an Aluminium Shape [6]

4.2 Wing Skin

Several options are studied for the manufacture of the wing. Standard manufacturing uses ribs as a shape to bend the skin on. However, this design only uses two of them at each extremity, forming a sort of monocoque. To ensure a good shape, ribs would need to be added, increasing material cost and overall weight. Another method would have been to create a folding external mould and use strong expanding foam to push the aluminium into place [19]. However, precise data concerning the pressure forces of expanding foam is hard to find especially knowing the strength of thick aluminium skin. Finally, the best method found is to press mould the wing into shape, in a similar way to the nose and the aft body of the fuselage. A simplified scheme of the manufacturing is shown in the next Figure.



Figure 16: Press Moulding of the Wing

4.3 Spars, Frames & Pylons

The spars are manufactured in a similar way to the fuselage skin, using extrusion. They are then riveted to the wing skin. Holes are made at the leading edge, spars and trailing edge to accommodate those. They are manufactured by rolling carbon fibre sheets with epoxy around a cylindrical mould and then cured in an autoclave oven. Those are finally glued with epoxy to cylindrical fittings which are then bolted into the spars, as described in the Wing Structural Design Report [20]. Similarly, heavy frames are extruded using the cross-section calculated using the load requirements found previously. The extrusion is then bent to form a circular shape and MIG welded to the fuselage skin.

4.4 Joints

Although the different parts are either glued or welded together, it is important to consider the use of additional joints to ensure that the stresses are correctly transmitted throughout the structure. Their precise shape and size calculations are not required in preliminary design. Between the spars and the skin, rivets are used due to their low price and effectiveness. Potentially, fittings can be used to increase the interaction surface. Between the wings and the fuselage, a wing box is bolted to the two heavy frames located at the spars of the wings to carry the loads. For waterproofing, silicon is used to fill the gaps between the main parts of the aircraft.

5 Materials

The UAV will be operating in a marine environment. It will spend a considerable amount of time hovering to drop the payload at its minimum flight altitude, 12 m. The exposed parts of the UAV will therefore be subject to water infiltration and corrosion. It is necessary to find a treatment solution against salt and corrosion to avoid the risk of damaging systems when operating offshore or in case of an emergency landing in water.

5.1 Material Treatment

Concerning the treatment, the aluminium surface needs to go through a chemical conversion. The first step is to remove the oxide coating naturally formed on the surface of aluminium alloys in contact with air. This is done using an acid deoxidizer such as Bonderite C-IC 33 Aero [21]. The surface is deep cleaned, and it is then treated using Alodine. That produces a coating offering corrosion resistance and paint adhesion without adding any extra weight. To better withstand the

harsh environment, paint is also applied to the exposed surfaces. The two types of paints which are often used in aircraft painting are epoxy or enamel-based [22]. On the one hand, epoxy is made of polyurethane, offering good adherence and chemical resistance and small oxidation and fading over time. It does not dry as hard as enamel and therefore does not chip or become brittle during its lifespan. On the other hand, the enamel is cheaper and less dangerous during application. Overall, due to the harsh environment, the UAV is operating in, epoxy is picked, with better capabilities and price not being an issue for such small surfaces.

5.2 Crash Protection

In theory, the UAV lands on an air cushion. However, in case of heavy winds or failed deployment, it might need to crash land. To protect the systems, the foam needs to offer good impact energy absorption, water resistance, low density and decent heat resistance when applied near hot systems. For the easiness of application, the foam needs to be self-expanding to fill up every gap possible. After a careful material selection, the most appealing choice is Polyurethane (PE) foam. It offers a good elongation (and energy absorption in consequence) and high self-expansion. PE foam is formed by the use of two separate components which are dispersed into to nozzle when applied. It is applied at the bottom of the fuselage, excluding the payload box. The predicted volume is 0.0062 m^3 , resulting in a 0.2% increase in overall weight. To quantify the effect of foam, the specific energy absorption [23] of PE foam is calculated with Equation 16. Values for Yield Strength σ_y and Max Strain ϵ are gathered using a typical PE foam stress vs. strain plot shown in Figure 29. The volume calculated results in a total energy absorption of 20.8 J. The total weight of avionics and actuators sums up to 2.44 kg. By equating the absorption energy of the PE foam to the kinetic energy induced by the systems, it can be deduced that the maximum speed of impact is 4.12 m/s. This calculation is not very accurate as the foam is not equally distributed between the different systems. Moreover, in case of a crash landing, the fuselage skin is likely to absorb most of the energy.

$$W = \sigma_y \epsilon$$

$$(16)$$

$$KE = \frac{1}{2} m V^2 \quad (17)$$

Heat shields are used for battery and engine so the foam is not subject to overheating. The use of foam within the wings is also studied, as a way to help the structure keep its aerodynamic shape. However, because the wing is already over-designed due to manufacturing constraints and also because it is not likely to impact the ground, this idea is not used in the final design. The weight increase would have been considerable with more than 700g added to the UAV.

5.3 Material Selection for Disposable Parts

Some parts of the design do not need to withstand the considerable aerodynamic loads. For instance, the payload doors, shown in Figure 19, are not directly connected to the rest of the monocoque. They only need to maintain their shape during flight. Similarly, as shown in Figure 31, the fairings are simple oval shapes fitted on the motors that reduce the aerodynamic drag. Those are not subject to high forces either. Moreover, the deck supporting the drone for take-off envelops the full fuselage and so has a low stress distribution. A representation of the deck is shown in Figure 32. For weight and price reasons, it is interesting to consider the use of polymer as they are strong enough to withstand reasonable loads and are much more cheap and easy to manufacture than aluminium or CFRP. Using the CES Granta software [24], a material selection is done optimising Young's Modulus, Fracture Resistance and Vickers's Hardness against Price per kg. The optimisations charts are shown in Figure 26, 27 and 28. From the plots, Polystyrene (PE) stands out from the other polymers. Using it allows for a weight reduction of 61% over an aluminium alloy. The main loads acting on the fairings are given by the Aerodynamics Team. Being extremely small, the fairings are manufactured with a 1 mm thickness PE skin. Similarly, the payload doors and take-off deck are manufactured with 1 mm thickness to fit the rest of the monocoque.

6 Secondary Structures

Although the majority of the work is focused on the design and sizing of the main components of the aircraft, it is important to consider the secondary structures that will be implemented in the finalised model. Some of them are presented in this section.

6.1 Camera Cutout

The first secondary structure to be designed is the camera cutout. The camera module is mounted on a stabilised gimbal, allowing it to rotate 180 degrees horizontally and 90 degrees vertically from the horizontal line towards the ground. To exploit the maximum potential of the camera, a canopy needs to be designated such that it does not obstruct the camera, even at maximum rotation. According to the spec sheet of the camera manufacturer [?], the thermal lens has a field of view (FOV) of 22 degrees horizontally and 18 degrees vertically. That allows determining the limits of the canopy on the nose of the fuselage. Figure 17 shows a scheme of the clearance needed for the camera. The resulting cutout appears on the final CAD model, shown in Figure 18

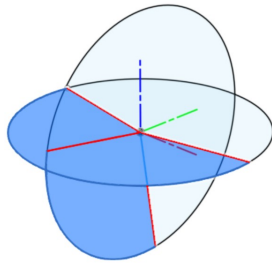


Figure 17: Camera Clearance

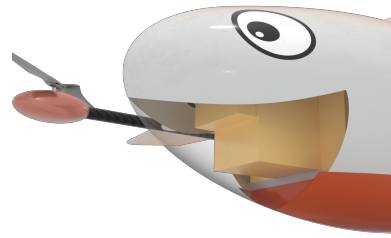


Figure 18: Camera Cutout on the CAD [3]

In blue is shown the clearance needed to take advantage of the full FOV of the thermal camera. This part needs to be perfectly see-through. In case of a scratch or crack apparition, it might impede the performance of the detection system which therefore needs to be changed. For easy maintenance, the part is sealed to an aluminium frame using silicon. The frame is then bolted to the rest of the fuselage. The canopy and the frame can be changed at the same time so the operator does not need to operate complex operations with silicon. The frame also allows for an increase in the overall strength of the canopy.

6.2 Payload Door

The payload door needs to open midair and close again by itself. Different methods are possible but the slide and close doors are the most efficient. Even though they require a bigger amount of motors to operate, they are more reliable than dropping doors which require a more complex closing mechanism. They also have less effect on the aerodynamics of the aircraft and can be entirely water-sealed. A simple CAD representation of the closing mechanism concept chosen is shown in Figure 19 and 20

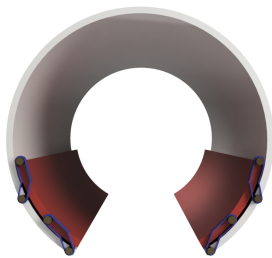


Figure 19: Payload Opening Mechanism

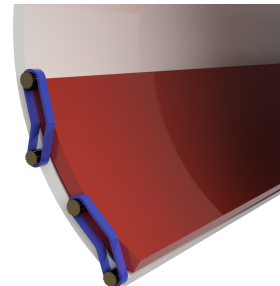


Figure 20: Payload Mechanism Close Up

7 Discussion

Even finalised, the project still has some design flaws. In future iterations, some specs can be changed to better tune the end product. For instance, throughout the design of the fuselage, it is clear that the constraints are mainly coming from manufacturing rather than the loads going through the structure. Finding an alternative going around the constraints might allow considerable weight savings. Even if the use of CFRP was scraped early in the design, it might however be interesting to consider hybrid structures, with the parts exposed to impacts made out of aluminium and the rest out of composite.

Although the fuselage has been carefully designed, it uses several assumptions that might induce inaccuracies in the results. To confirm the calculations, it would be relevant to do a Finite Element Analysis (FEA) of the loads along the fuselage, especially in the case of a hybrid structure used where classic analysis is not applicable.

Furthermore, Heavy Frames are designed using a specific high load case. However, other scenarios with multiple coupled forces might give surprising results. For instance, when handled in specific ways before take-off, the drone might go through loads that were predicted initially, even more so if the drone gets dropped, handled in the wrong position and so on. However, those loads are complicated to model and an in-depth analysis is necessary to get accurate results. In consequence, Heavy Frames design could take advantage of FEA and be even more optimised, potentially by using generative design.

Currently, the manufacturing processes are simply cited and explained. That does not include a feasibility study of each idea. Doing so would induce a better understanding of the different pros and cons of each solution to converge to a more viable manufacturing procedure. Moreover, the price calculations for the different methods are approximate, making the choices even more difficult. Better price estimations would then give hints about the amount of UAV needed to make the project viable, by comparing the price of labour of handmade structures against high manufacturing overhead costs such as the moulds.

In preliminary design, secondary structures are only defined and explained but not sized. In consequence, there is potential interference or overlap between the systems placement and the secondary structures. After an appropriate sizing, a precise drawing of the placement of the different systems could be done, taking into account the position of the secondary structures. Different attachment points to the structure could then be designed, allowing to optimise the placement of the foam to offer the best protection to the sensible parts of the UAV.

8 Conclusion

Overall, the different teams have collaborated to end up with a fully sized aircraft. Collaboration is sometimes hard when some changes are made by a member but not acknowledged by the rest of the group. Design freezes allow mitigating this. Currently, the design has some drawbacks with several constraints hindering the global performance of the aircraft. The concept essentially went through 3 sizing iterations. In a real-life scenario, much more iterations are necessary, probably refining the characteristics to produce a more capable product. Concerning the structure, even with a rigorous load analysis, the elaborate sizing gets perturbed by the manufacturing constraints, leading to an over-designed structure, capable of withstanding considerable loads. Still, it is not weight-optimised, the main goal of an aircraft structure. Nonetheless, this over-sizing leads to a slick design, making the manufacturing cheaper. A more hybrid concept, taking advantage of both aluminium and composite, would probably make the structure much more efficient. The rest of the report focused on different aspects of the structure design such as the material selection, the manufacturing processes and the implementation of secondary structures. Those parts went less in-depth than the actual fuselage and will have to be refined further down the project.

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Appendix

9 Literature Review

This table shows the different hypothermia risks depending on the temperature of the water. It is used to determine the maximum time to target allowable.

Water Temperature in Degrees F (Degrees C)	Loss of Dexterity (with no protective clothing)	Exhaustion or Unconsciousness	Expected Time of Survival	Recommended Paddling Clothing*
32.5 (0.3)	Under 2 min.	Under 15 min.	Under 15 to 45 min.	Drysuit with fleece layers, neoprene booties, beanie, gloves
32.5 to 40 (0.3 to 4.5)	Under 3 min.	15 to 30 min.	30 to 90 min.	Drysuit with fleece layers, neoprene booties, beanie, gloves
40 to 50 (4.5 to 10)	Under 5 min.	30 to 60 min.	1 to 3 hrs.	Drysuit with fleece or polypro layers, neoprene booties, beanie, gloves
50 to 60 (10 to 15.5)	10 to 15 min.	1 to 2 hrs.	1 to 6 hrs.	Drysuit with polypro layers or wetsuit plus drytop, neoprene booties, hat
60 to 70 (15.5 to 21)	30 to 40 min.	2 to 7 hrs.	2 to 40 hrs.	Wetsuit with light paddling jacket, paddling shoes, hat
70 to 80 (21 to 26.5)	1 to 2 hrs.	2 to 12 hrs.	3 hrs. to indefinite	Thin or shorty wetsuit
Over 80 (Over 26.5)	2 to 12 hrs.	Indefinite	Indefinite	Swimsuit

Figure 21: Hypothermia Risks depending on the Water Temperature [7]

This figure is a map surveying the different migrants incidents. It is used to approximate the best range possible for the mission profile.

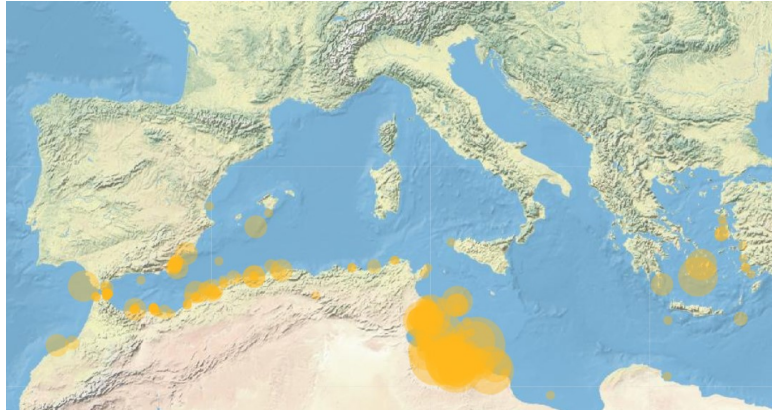


Figure 22: Migrant Incidents On a Map [8]

This graph shows the average number of migrants per boat in the Channel in 2020/2021. It allows to calculate the ideal number of buoys that the UAV should carry.

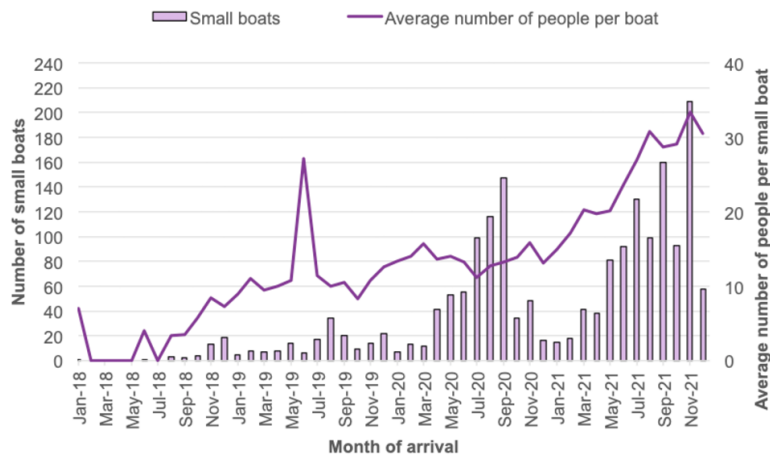


Figure 23: Number of Migrants Per Boat surveyed in the Channel [9]

10 Fuselage Design

This plot shows the load distribution along the fuselage, including the tail load and the reaction forces at the spars.

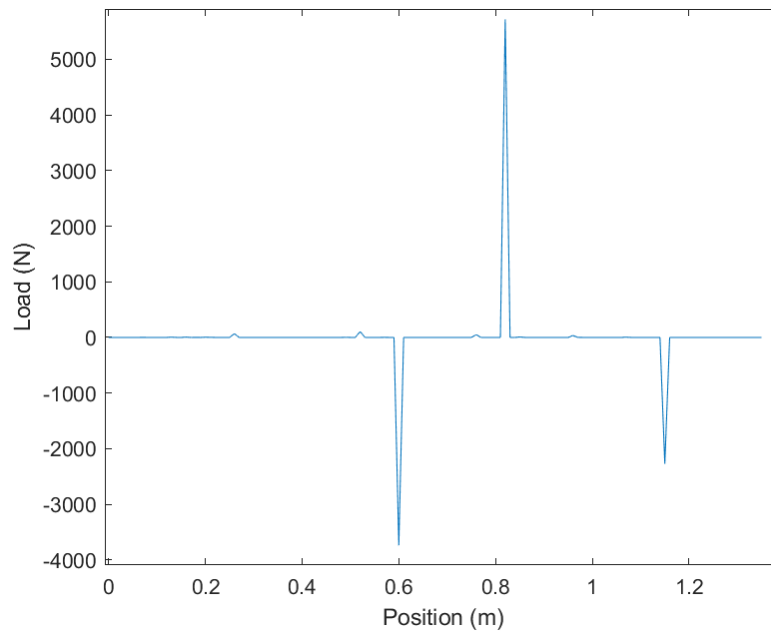


Figure 24: Load Distribution along the Fuselage

This plot shows the result of experimental data for buckling of different cylinders with variable length, radius and thickness. It gives a correction factor for a more accurate buckling stress than the theoretical buckling stress of a beam under axial loading.

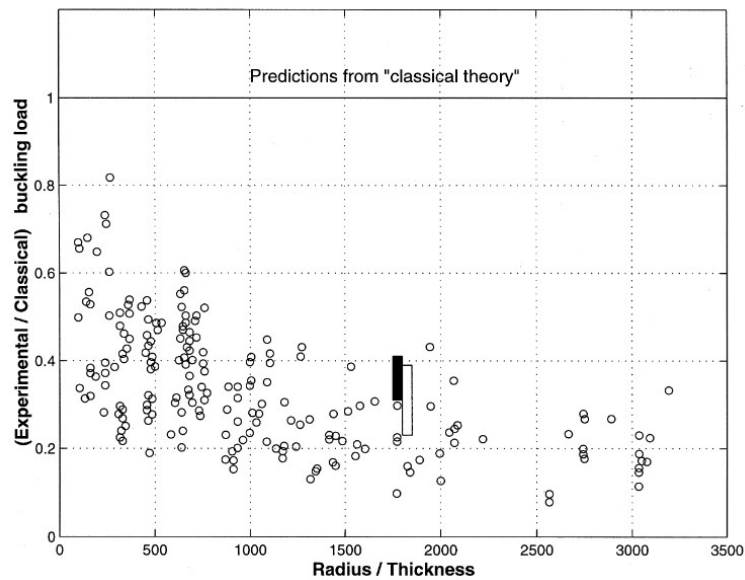


Figure 25: Experimental Data for Cylinder Buckling under Axial Loading [10]

11 Materials

This plots gives the best material for Young's Modulus , Fracture Toughness and Water and Salt resistance against Price per kg.

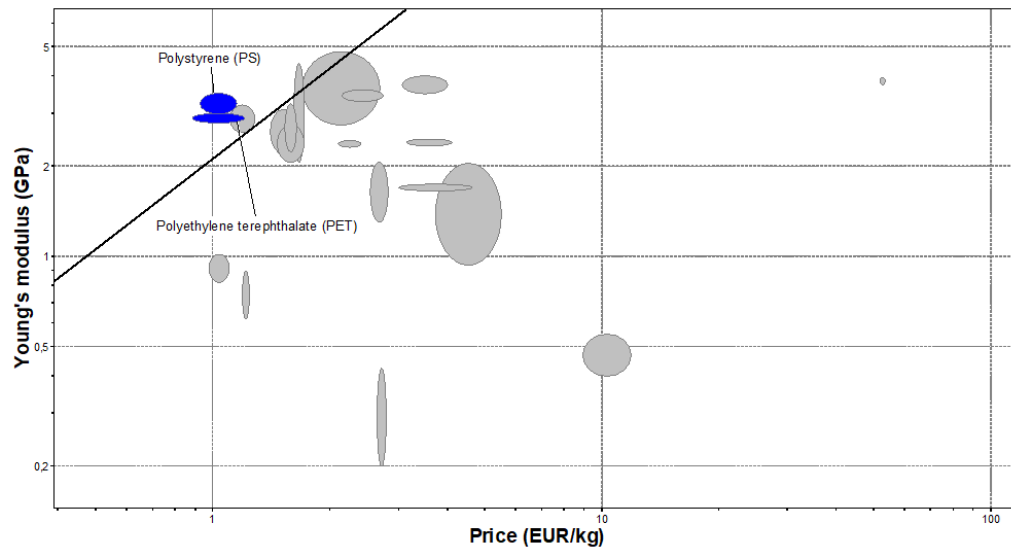


Figure 26: Young's Modulus vs. Price per kg plot

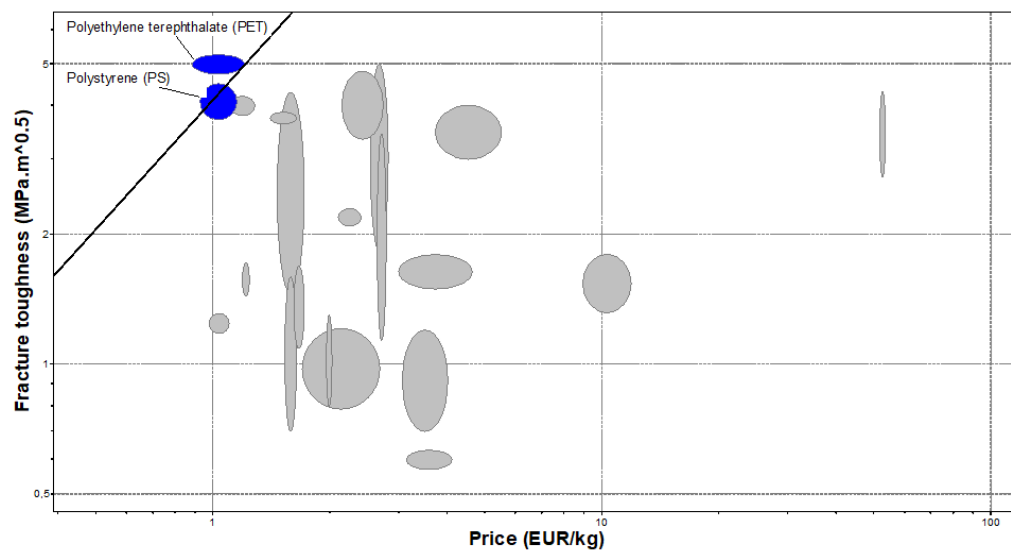


Figure 27: Fracture Toughness vs. Price per kg plot

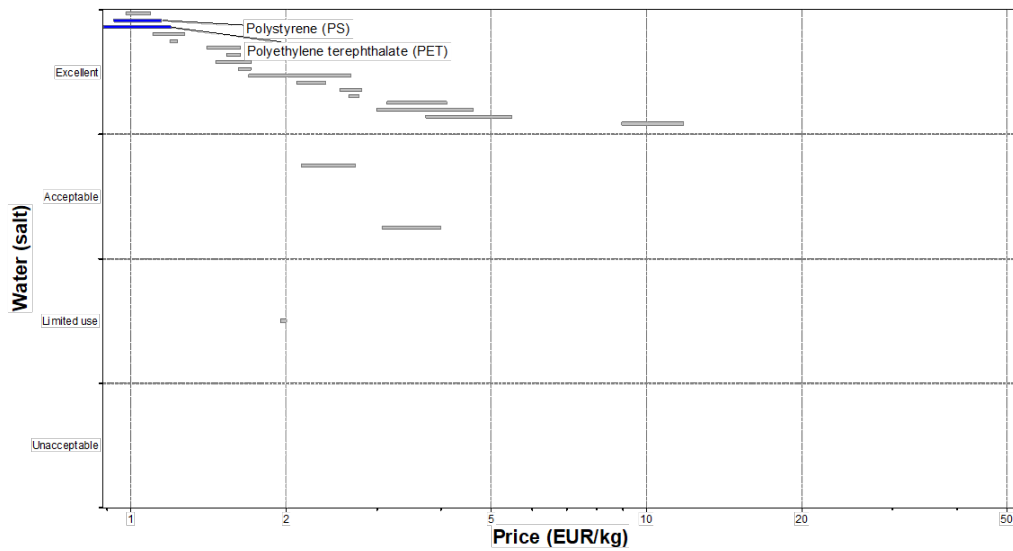


Figure 28: Water And Salt Resistance vs. Price per kg plot

This plot is a Stress vs. Strain curve for polyurethane foam. It is used to derive the energy absorption of the foam. [23].

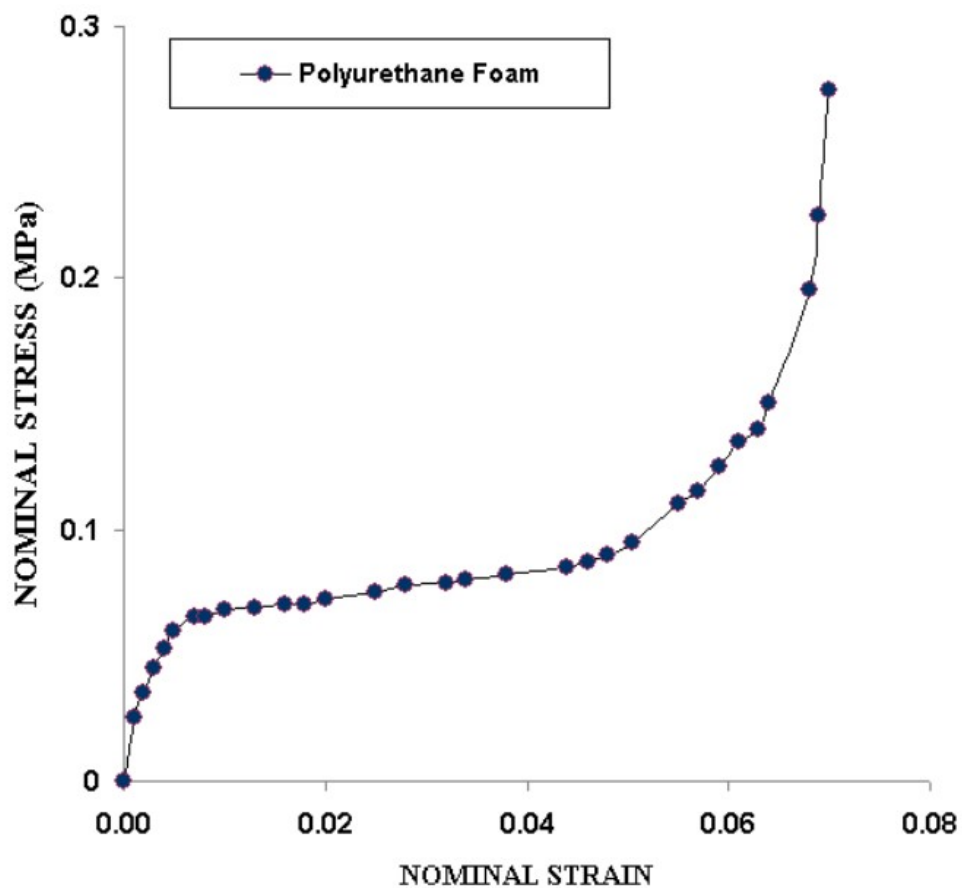


Figure 29: Typical PE Foam Stress vs. Strain Curve [11]

This table summarises the different characteristics of materials used for the structure design.

Component	Material	ρ (kg/m ³)	E (GPa)	Price (£/kg)	σ_Y (MPa)	σ_C (MPa)	σ_T (MPa)	K_{IC}
Canopy	Cleartran (ZnS)	4090	74.5	-	-	60	60	1
Skin	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Spars/Ribs	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Frames	Aluminium 7055 T74511	2850	72.7	4.29	581.5	567.5	627	36.15
Pylons	Epoxy/HS CF, resin infused woven fabric, biaxial layup	1515	61	19	653.5	710	653.5	31.2

Figure 30: Proprieties of the Different Materials used in the Design [12]

This is a CAD representation of the fairings used to reduce aerodynamic perturbations on the wings and drag. It is made out of Polystyrene.

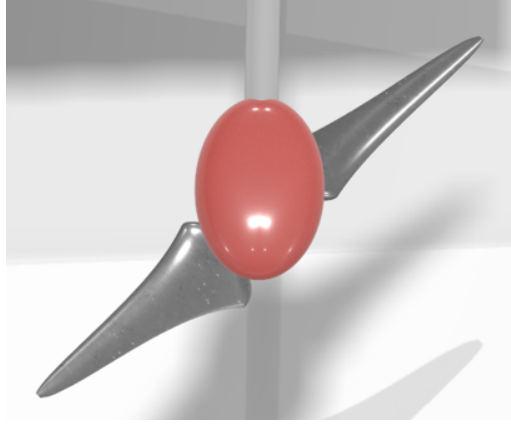


Figure 31: Fairings Made of Polystyrene Polymer [3]

This is a CAD representation of the take-off deck used to ensure a safe take off by avoiding any rotation of the drone on the ground. It is made out of Polystyrene.

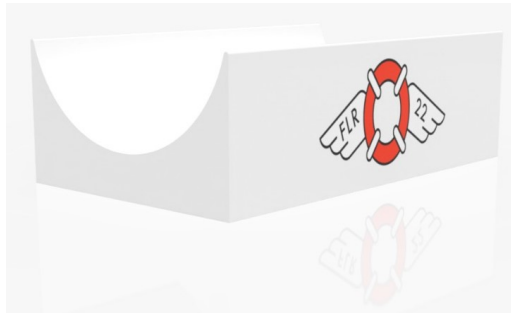


Figure 32: Take-Off Deck Made of Polystyrene Polymer [3]

12 MATLAB Codes Used

This are all the MATLAB codes used for the structural design of the fuselage.

Shear Force And Bending Moment Calculations

```
clear  
clc  
close all
```

% Initial Values

```
radius= 0.176;  
diam=2*radius;  
n=4.5
```

```
num = readcell('differentweights2.xlsx'); %reading values from excel  
weight=cell2mat(num(:,2))*10^-3; %systems load  
x_sys=cell2mat(num(:,3))*10^-3; %systems x position  
name=num(:,1); %naming (in case needed)
```

```
load1=weight*9.81; %transforming to N  
tailload=-2274 %cell2mat(num(1:2,9)); %Getting the tail load in N  
x_tail=1.2-0.25*0.2116; %tail position
```

%Position of spars

```
chord=cell2mat(num(2,12));  
meanchord=cell2mat(num(1,12));
```

```
x_acw=cell2mat(num(3,12));  
x_wing=x_acw-0.25*meanchord;
```

```
front_spar = round(x_wing+0.15*chord,2);  
rear_spar = round(x_wing+0.7*chord,2);
```

%Setting Up Load Case

```
mat=[x_sys, weight*9.81;x_tail,tailload(1)]; %force matrix  
pos = size(mat,1);  
tot_mass = sum(mat(:,2)); %total mass  
moment = 0;
```

```
% Summation of the moments
```

```
for i = 1:pos  
moment = moment + mat(i,1)*mat(i,2);  
end
```

```
% Solving for the reaction forces at the spars
```

```
A = [front_spar, -rear_spar; 1, -1];  
B = [moment; tot_mass];  
reac = linsolve(A,B);
```

```
%Setting up the arrays
```

```
internal.position = [0:0.01:1.35]';  
internal.load = zeros(length(internal.position),1);
```

```
% Repartition of the fuselage over the different stations
```

```
for i = 1:length(internal.position)  
internal.load(i) = (4.6740*9.81)/length(internal.position);
```

```
% Allocation of the system loads at the right station
```

```
for j = 1:pos  
if round(mat(j,1),2) == round(internal.position(i),2)  
if j == 1.35
```

```
break
```

```
end
```

```
internal.load(i) = internal.load(i)+mat(j,2);
```

```
end
```

```
end
```

```
%Finding the position of the spars on the position array
```

```
reaction_pos_1 = 61; %find(internal.position==front_spar);  
reaction_pos_2 = 83; %find(internal.position==rear_spar);
```

```
%Inputing reaction forces at spars
```

```
internal.load(reaction_pos_1) = -reac(1);  
internal.load(reaction_pos_2) = reac(2);
```

```
% Set up of the arrays
```

```
internal.sf(1)=0;  
internal.bm(1)=0;  
internal.position(1)=0;
```

```
for k = 2:length(internal.position)
```

```
internal.sf(k) = internal.sf(k-1)+internal.load(k)*n;  
internal.bm(k) = internal.bm(k-1)+(internal.position(k)-  
internal.position(k-1))*internal.sf(k);
```

```
end
```

```
end
```

```
maxsf = max(abs(internal.sf))  
maxbm = max(abs(internal.bm))
```

```
%Setting Up Load Case 2
```

```
mat=[x_sys, weight*9.81;x_tail,336.27]; %force matrix  
tot_mass = sum(mat(:,2)); %total mass  
moment = 0;
```

```
% Summation of the moments
```

```
for i = 1:pos
```

```
moment = moment + mat(i,1)*mat(i,2);
```

```
end
```



```
% Solving for the reaction forces at the spars
```

```
A = [front_spar, -rear_spar; 1, -1];  
B = [moment; tot_mass];  
reac = linsolve(A,B);
```

```
%Setting up the arrays
```

```
internal.position = [0:0.01:1.35]';  
internal.load2 = zeros(length(internal.position),1);
```

```
% Repartition of the fuselage loads over the different stations
```

```
for i = 1:length(internal.position)  
  
internal.load2(i) = (4.6740*9.81)/length(internal.position);
```

```
% Allocation of the system loads at the right station
```

```
for j = 1:pos  
  
if round(mat(j,1),2) == round(internal.position(i),2)  
  
if j == 1.35  
  
break  
  
end  
  
internal.load2(i) = internal.load2(i)+mat(j,2);  
  
end  
  
end
```

```
%Finding the position of the spars on the position array
```

```
reaction_pos_1 = 61; %find(internal.position==front_spar);  
reaction_pos_2 = 83; %find(internal.position==rear_spar);
```

```
%Inputing reaction forces at spars
```

```

internal.load2(reaction_pos_1) = -reac(1);
internal.load2(reaction_pos_2) = reac(2);

% Set up of the arrays

internal.sf2(1)=0;
internal.bm2(1)=0;
internal.position(1)=0;

for k = 2:length(internal.position)

internal.sf2(k) = internal.sf2(k-1)+internal.load2(k)*n;
internal.bm2(k) = internal.bm2(k-1)+(internal.position(k)-
internal.position(k-1))*internal.sf2(k);

end

end

figure
plot(internal.position,internal.load)
xlabel('Position (m)');
ylabel('Load (N)');

figure;
plot(internal.position,internal.sf);
hold on
plot(internal.position,internal.sf2);
xlabel('Position (m)');
ylabel('Shear Force (N)');
legend('Manoeuvre', 'Dive')

figure;
plot(internal.position,internal.bm)
hold on
plot(internal.position,internal.bm2)
xlabel('Position (m)');
ylabel('Bending Moment (Nm)');
legend('Manoeuvre', 'Dive')
shearmodulus=320*10^6;

shearflow=maxsf/(pi*diameter); %max shear flow

thickness=1.5*shearflow/shearmodulus %minimum thickness required

```

Buckling Calculations

```
tmin=0.001; % Thickness chosen  
diam=0.352
```

```
b=((diam*pi)/1); %circumference
```

```
E=72.7*10^9; %youngs modulus  
poisson=0.33; %poisson ratio
```

```
% Max Values from ESDU
```

```
theoreticalmax= (E/sqrt(3*(1-poisson)))*(tmin/radius)  
experimentalmax=0.5*theoreticalmax
```

```
% Max reached
```

```
icross= (pi*radius^4)/4-(pi*(radius-tmin)^4)/4; % second moment of  
inertia of hollow cylinder  
sigmab=(maxbm*radius)/icross %max stress
```

```
% Ratio of max in this case to max experimental
```

```
ratio=sigmab/experimentalmax
```

Weight Comparison For two sets of fuselage

%Fuselage 1

```
length=120;  
radius=17.7;  
density=2.71;  
innerradius=17.6;
```

```
vol=pi*(radius^2-innerradius^2)*length;  
weight=vol*density/(10^3)  
circ=2*pi*radius;  
wet=circ*length;
```

%Fuselage 2

```
length2=135;  
radius2=14.4;  
density2=2.71;  
innerradius2=14.2;
```

```
vol2=pi*(radius2^2-innerradius2^2)*length2;  
weight2=vol2*density2/(10^3);  
circ2=2*pi*radius2;  
wet2=circ2*length2;
```

Wise Curve Force Calculations

```
clc
clear
close all
```

```
D=0.352; %diameter of fuselage
tensile_stress=675*10^6; % tensile yield stress
shear_stress=675*10^6/sqrt(3); % shear yield stress
```

```
%Wing
% F = 1.0875e03 ; %root shear force divided by two because of spars
% M = 454.05/1; %bending moment
```

```
F = 1.1554e03 ; %root shear force divided by two because of spars
M = 186.0387; %bending moment
```

```
thet=0; %position of the component
```

```
%equations
```

```
%2nd wing load
```

```
P=F*cosd(thet);
Q=F*sind(thet);
T=M;
```

```
phi=[0:10:360]; % in degrees
radphi=phi.*(pi/180); % in rad
R=D/2; %radius of fuselage
```

```
%Tangential load
```

```
N_t1=(P/2/pi).*((pi-radphi).*cos(radphi)-(0.5.*sin(radphi)));
```

```

S_t1=(P/2/pi).*(1+(0.5.*cos(radphi))-((pi-radphi).*sin(radphi)));
M_t1=(P*R/2/pi).*(((pi-radphi).*(1-cos(radphi)))-
(1.5.*sin(radphi)));

```

%Radial load

```

N_r1=(Q/2/pi).*((1.5.*cos(radphi))+(pi-radphi).*sin(radphi));
S_r1=(Q/2/pi).*(((pi-radphi).*cos(radphi))-(0.5.*sin(radphi)));
M_r1=(Q*R/2/pi).*(1+(0.5.*cos(radphi))-((pi-radphi).*sin(radphi)));

```

%Moment load

```

N_m1=(T/pi/R).*(sin(radphi));
S_m1=(T/2/pi/R).*(1+2.*cos(radphi));
M_m1=(T/2/pi).*((pi-radphi)-(2.*sin(radphi)));

```

```

N_r = N_m1+N_r1+N_t1;
S_r = S_m1+S_r1+S_t1;
M_r = M_m1+M_r1+M_t1;

```

%2nd wing load

```

P=F*cosd(180-thet);
Q=F*sind(thet);
T=M;

```

%Tangential load

```

phi=[0:10:360]; %in degrees
radphi=phi.*(pi/180); %in rad

```

```

N_t2=(P/2/pi).*((pi-radphi).*cos(radphi)-(0.5.*sin(radphi)));
S_t2=(P/2/pi).*(1+(0.5.*cos(radphi))-((pi-radphi).*sin(radphi)));
M_t2=(P*R/2/pi).*(((pi-radphi).*(1-cos(radphi)))-
(1.5.*sin(radphi)));

```

%Radial load

```

N_r2=(Q/2/pi).*((1.5.*cos(radphi))+(pi-radphi).*sin(radphi));
S_r2=(Q/2/pi).*(((pi-radphi).*cos(radphi))-(0.5.*sin(radphi)));
M_r2=(Q*R/2/pi).*(1+(0.5.*cos(radphi))-((pi-radphi).*sin(radphi)));

```

```
%Moment load
```

```
N_m2=(T/pi/R).*(sin(radphi));
```

```
S_m2=(T/2/pi/R).*(1+2.*cos(radphi));
```

```
M_m2=(T/2/pi).*((pi-radphi)-(2.*sin(radphi)));
```

```
N_l = N_m2+N_r2+N_t2;
```

```
S_l = S_m2+S_r2+S_t2;
```

```
M_l = M_m2+M_r2+M_t2;
```

```
% Sum of the loads
```

```
M=M_l+M_r;
```

```
N=N_l+N_r;
```

```
S=S_l+S_r;
```

```
%Plots
```

```
figure;
```

```
plot(phi,N);
```

```
hold on;
```

```
plot(phi,S);
```

```
hold on;
```

```
plot(phi,M);
```

```
xlabel('\theta (deg)');
```

```
ylabel('Force (N) and Moment (Nm)');
```

```
legend('Axial Force N','Shear Force S','Moment M');
```

```
maxN = max(N);
```

```
maxS = max(S);
```

```
maxM = max(M);
```

```
A_a = maxN/tensile_stress %axial stress
```

```
A_s = maxS/shear_stress %perpendicular stress
```

I Beam optimisation using a loop

```
count=1
for tmin=0.001:0.001:0.01 %thickness variation

    for H=(0.001:0.001:0.1) %H variation
        for B=(0.001:0.001:0.1) %B variation

            A=(H+2*tmin)*(B+tmin)-2*H*((B+tmin)-tmin)/2;
            I_req = maxM*H/(2*tensile_stress);
            inertia= (B*H^3)/12-(B-tmin)*(H-2*tmin)*((H*tmin)^3)/12;
            if inertia>I_req

                Hg(count,1)=H;
                Hg(count,2)=B+tmin;
                Hg(count,3)=tmin;
                Hg(count,4)=A;
                Hg(count,5)=inertia;
                Hg(count,6)=I_req;
                count=count+1;

            end
        end
    end
end

% Displaying optimal values

area=Hg(:,4);
[M2,I2]=min(area)
```


Weight estimation for Heavy Frames

```
H=final(1)
B=final(2)
t=final(3)
density=2710
```

```
rout1=D/2 %external radius
```

```
rin1=rout1-B %internal radius
```

```
volumerectangle=pi*(H+2*t)*(rout1^2-rin1^2) %rectangle heavy frame
volume
```

```
rin2=rout1-(B-t)/2
```

```
volume2=pi*H*(rout1^2-rin2^2) %first part to take of volume for I
shape
```

```
rout3=rin1+(B-t)/2
```

```
volume3=pi*H*(rout3^2-rin1^2) %second part to take of volume for I
shape
```

```
volumefinal=volumerectangle- volume2 - volume3 %final volume
```

```
weight= density*volumefinal
```